

Table of contents

Potential (intrinsic energy of inertia)

Electric Potential U :

$$e = \text{electron charge} \quad q = ne = \text{charge} \quad n \in \mathbb{N} \quad (1)$$

$$U = \frac{E_{pot}}{q} = \frac{m}{q}c^2 = \kappa c^2 \quad (2)$$

Energy density electric:

$$w_e = \frac{1}{2}\epsilon_0 \mathcal{E}^2 \quad (3)$$

Energy density magnetic:

$$w_m = \frac{1}{2} \frac{1}{\mu_0} \mathcal{B}^2 \quad (4)$$

Electric Energy:

$$W_e = \frac{1}{2}CU^2 \quad (5)$$

Magnetic Energy:

$$W_m = \frac{1}{2}LI^2 \quad (6)$$

Power:

$$P = VI = (U_1 - U_2) \frac{Q}{t} \equiv \frac{\text{Work}}{\text{Time}} = \text{Force} \cdot \text{Velocity} = \frac{\text{Energy difference}}{\text{Time}} \quad (7)$$

Conductor (charge conductive element for energy transfer) condition:

$$\sigma \cdot t \gg \epsilon_0 \quad (8)$$

Slow changing current (quasi-stationary, low frequencies) condition:

$$\frac{v}{c} \ll \frac{1}{c} \frac{\mathcal{E}}{\mathcal{B}} \ll \frac{c}{v} \quad (9)$$

Energy conservation for (large extended) conductive medium:

$$\int \left(\frac{\partial}{\partial t} (we + w_m) + \cdot \mathcal{S} + j \cdot \mathcal{E} \right) dr = \frac{d}{dt} W_e + \frac{d}{dt} W_m + D_S + \int (j \cdot \mathcal{E}) dr \quad (10)$$
$$\stackrel{\text{low frequency}}{=} \frac{1}{2} \frac{d}{dt} (LI^2) + \frac{1}{2} \frac{d}{dt} \left(\frac{Q^2}{C} \right) + (RI^2 - VI) = 0$$

Energy of thin conductor (concentrated conductive medium without electric charge in its surrounding), Voltage (potential difference, tension):

$$V = U_1 - U_2 = r\mathcal{E} = L\dot{I} + RI + \frac{Q}{C} = \dot{\Psi} + RI + \frac{It}{C} \quad (11)$$

V = Potential difference I = Current R = Resistance

L = Inductance C = Capacitance $Q = \text{elec. charge}$ $\Psi = LI = \text{mag. Flux}$

$\mathcal{S} = \mathcal{E} \times \mathcal{H} = \text{Poynting-Vector} = \text{intensity of radiation}$

$D_S = \int (\cdot \mathcal{S}) dr = \int \text{div}(\mathcal{E} \times \mathcal{H}) dr = \text{dissipation energy lost from radiation} \approx 0$ for low frequencies ω

$d = \sqrt{\frac{1}{\mu\sigma\omega}} = \text{skin depth (surface) of conductor}$ $r = \frac{d}{\sqrt{2}i}\rho = \frac{d}{1+i}\rho = \text{radius (extension) of conductor}$

$\rho = kr = \text{"radius of curvature } k"$ $\omega = 2\pi\nu = k\lambda\nu = \frac{1}{\sqrt{LC}} = \text{frequency}$

$\mu = \text{permeability of conductor}$ $\sigma = \text{conductivity of conductor}$

Low Frequency:

$$\begin{aligned} \omega \ll \frac{1}{\mu\sigma} \frac{2}{r^2} &\iff r \ll d \iff \rho \ll 1 \\ L \approx \text{const.} \quad C \approx \text{const.} \quad D_S &\approx 0 \end{aligned} \quad (12)$$

High Frequency:

$$\omega \gg \frac{1}{\mu\sigma} \frac{2}{r^2} \iff r \gg d \iff \rho \gg 1 \quad (13)$$

$$L(t) \neq \text{const.} \quad C(t) \neq \text{const.} \quad D_S \propto \omega^4 \gg 0$$

Energy:

$$E_{kin}^{max} = eV_0 = h(\nu - \nu_0) \quad (14)$$

Action Quantum, Planck Constant (Energy-Frequency-Slope)¹:

$$\begin{aligned} h = \tan(\alpha) &= \frac{E_{kin}}{\nu_i - \nu_0} = \frac{eV_{0i}}{\nu_i - \nu_0} \equiv \\ \frac{\Delta E}{\nu_0} &= \frac{W}{\nu_0} = \frac{F_{EM} \cdot x}{\nu_0} = \hbar k \lambda \end{aligned} \quad (15)$$

$$\hbar = \frac{\Delta E}{\omega} = \quad (16)$$

$$\begin{aligned} \alpha &= \angle(E_{kin}, \nu) \quad e = \text{electron charge} \\ V_0 &= \text{Opposing Potential } (I = 0) \quad E_{kin}^{max} = eV_0 \end{aligned} \quad (17)$$

¹Wirkungsquantum